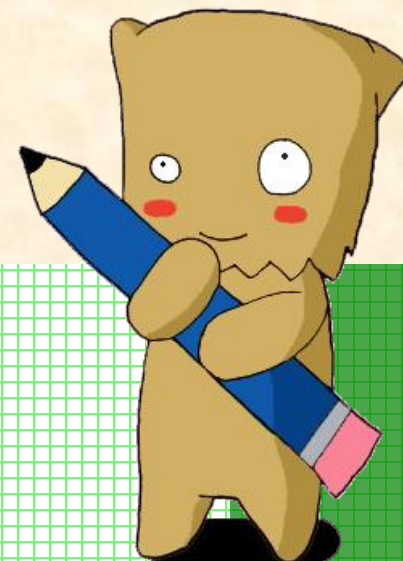




Chapter 4

DECISION MAKING FOR A SINGLE SAMPLE



Tanasanee Phienthrakul

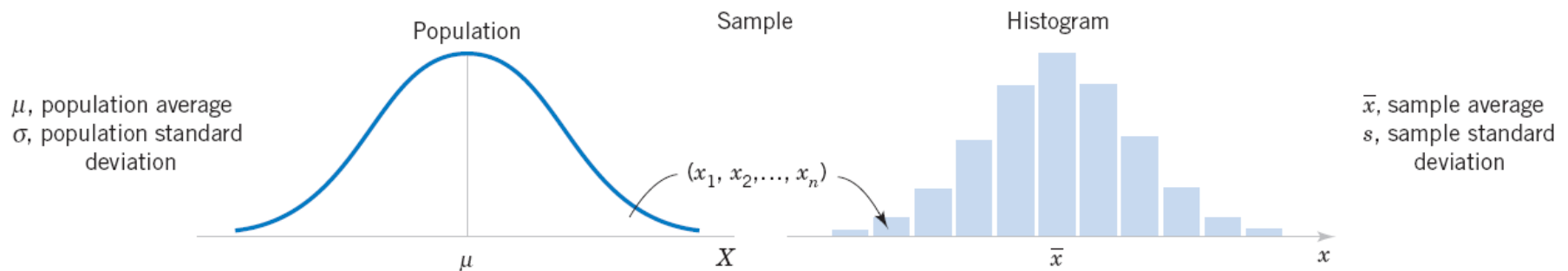
Decision Making for a Single Sample



- Statistical Inference and Point Estimation
- Inference on the Mean of a Population, Variance Known
- Inference on the Mean of a Population, Variance Unknown
- Inference on the Variance of a Normal Population
- Inference on a Population Proportion
- Hypothesis Testing

Statistical Inference

- The field of statistical inference consists of those methods used to make decisions or draw conclusions about a **population**.
- These methods utilize the information contained in a **sample** from the population in drawing conclusions.
- Example: Estimate the average height of the class.



Relationship between Population and Sample

Point Estimation

- A point estimate of some population parameter θ is a single numerical value $\hat{\theta}$ of a statistic $\hat{\Theta}$.

Unknown Parameter θ	Statistic $\hat{\Theta}$	Point Estimate $\hat{\theta}$
μ	$\bar{X} = \frac{\sum X_i}{n}$	$\bar{x}$
σ^2	$S^2 = \frac{\sum (X_i - \bar{X})^2}{n - 1}$	s^2
p	$\hat{P} = \frac{X}{n}$	$\hat{p}$
$\mu_1 - \mu_2$	$\bar{X}_1 - \bar{X}_2 = \frac{\sum X_{1i}}{n_1} - \frac{\sum X_{2i}}{n_2}$	$\bar{x}_1 - \bar{x}_2$
$p_1 - p_2$	$\hat{P}_1 - \hat{P}_2 = \frac{X_1}{n_1} - \frac{X_2}{n_2}$	$\hat{p}_1 - \hat{p}_2$

Point Estimation

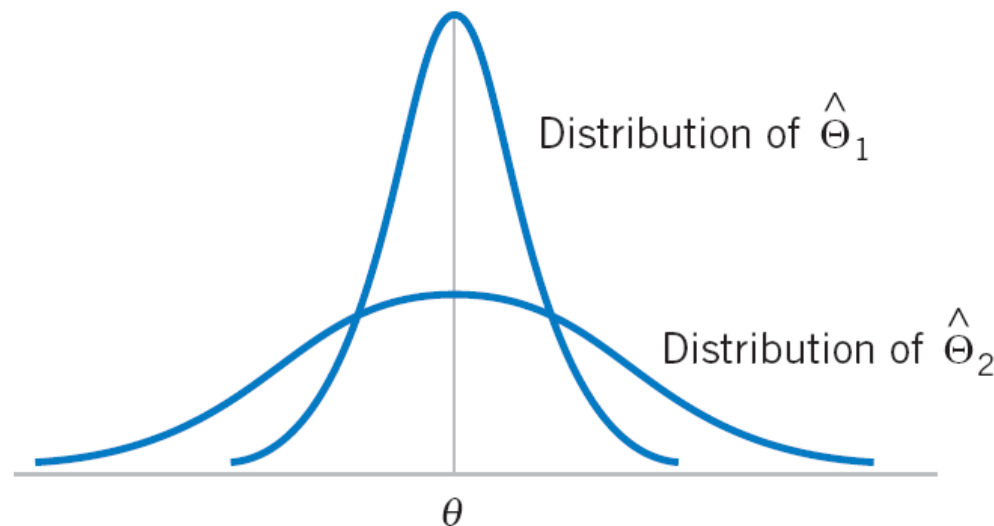
- An estimator should be close in some sense to the true value of the unknown parameter.
- $\hat{\theta}$ is an **unbiased estimator** of θ if $E(\hat{\theta}) = \theta$
- If the estimator $\hat{\theta}$ is not unbiased, then the difference $E(\hat{\theta}) - \theta$ is called the **bias of the estimator $\hat{\theta}$**

Different Unbiased Estimators

- Sometimes there are several unbiased estimators of the sample population parameter.
- **Example:** a random sample of size $n = 10$.
 - Sample mean
 - Sample median
 - The first observation X_1
- All are unbiased estimator of population X .
- Cannot rely on the property of unbiasedness alone to select the estimator.

Different Unbiased Estimators

- Suppose that Θ_1 and Θ_2 are unbiased estimators of θ
- The variances of these two distribution of each estimator may be different.
 - Θ_1 has a smaller variance than Θ_2 does.
 - Θ_1 is more likely to produce an estimate close to the true value θ .



Exercise

- Let $X_1, X_2, \dots, X_9$ denote a random sample from a population having mean μ and variance σ^2 . Consider the following estimators of μ :

A. $\Theta_1 = (X_1 + X_2 + \dots + X_9)/9$

B. $\Theta_2 = (3X_1 - X_6 + 2X_4)/2$

C. $\Theta_3 = (5X_1 - X_6 - 2X_4)/2$

- Is either estimator unbiased?
- Which estimator is better? In what sense is it better?

Decision Making for a Single Sample



- Statistical Inference and Point Estimation
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- Hypothesis Testing

Interval estimation

- In the previous topic we considered how a parameter can be estimated from sample data. However, it is important to understand how good is the estimate obtained.
- Bounds that represent an interval of plausible values for a parameter are an example of an **interval estimate**.

Confidence Interval on the Mean of a Normal Distribution, Variance Known

$\bar{X}$ is normally distributed with mean μ and variance σ^2/n . We may standardize $\bar{X}$ by subtracting the mean and dividing by the standard deviation, which results in the variable

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

The random variable Z has a standard normal distribution.

Confidence Interval on the Mean of a Normal Distribution, Variance Known

A **confidence interval** estimate for μ is an interval of the form $l \leq \mu \leq u$, where the end-points l and u are computed from the sample data. Because different samples will produce different values of l and u , these end-points are values of random variables L and U , respectively. Suppose that we can determine values of L and U such that the following probability statement is true:

$$P\{L \leq \mu \leq U\} = 1 - \alpha$$

where $0 \leq \alpha \leq 1$. There is a probability of $1 - \alpha$ of selecting a sample for which the CI will contain the true value of μ . Once we have selected the sample, so that $X_1 = x_1, X_2 = x_2, \dots, X_n = x_n$, and computed l and u , the resulting **confidence interval** for μ is

$$l \leq \mu \leq u$$

Confidence Interval on the Mean of a Normal Distribution, Variance Known

- Development of the Confidence Interval and its Basic Properties
 - The endpoints or bounds l and u are called **lower-** and **upper-confidence limits**, respectively.
 - Since Z follows a standard normal distribution, we can write:

$$P\left\{-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right\} = 1 - \alpha$$

Now manipulate the quantities inside the brackets by (1) multiplying through by $\sigma/\sqrt{n}$, (2) subtracting $\bar{X}$ from each term, and (3) multiplying through by -1 . This results in

$$P\left\{\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha$$

Confidence Interval on the Mean of a Normal Distribution, Variance Known

- Development of the Confidence Interval and its Basic Properties
- Definition

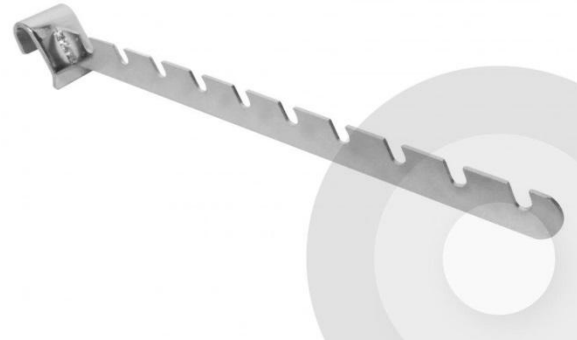
If $\bar{x}$ is the sample mean of a random sample of size n from a normal population with known variance σ^2 , a $100(1 - \alpha)\%$ CI on μ is given by

$$\bar{x} - z_{\alpha/2}\sigma/\sqrt{n} \leq \mu \leq \bar{x} + z_{\alpha/2}\sigma/\sqrt{n}$$

where $z_{\alpha/2}$ is the upper $100\alpha/2$ percentage point of the standard normal distribution.

Example: Metallic Material Transition

- ASTM Standard E23 defines standard test methods for notched bar impact testing of metallic materials.
- Ten measurements of impact energy (J) on specimens of A238 steel cut at 60°C are as follows: 64.1, 64.7, 64.5, 64.6, 64.5, 64.3, 64.6, 64.8, 64.2, and 64.3.
- Assume that impact energy is normally distributed with $\sigma = 1 J$. We want to find a 95% CI for μ , the mean impact energy.
- The required quantities are $z_{\alpha/2} = z_{0.025} = 1.96$, $n = 10$, $\sigma = 1$, and $\bar{x} = 64.46$.



Example: Metallic Material Transition

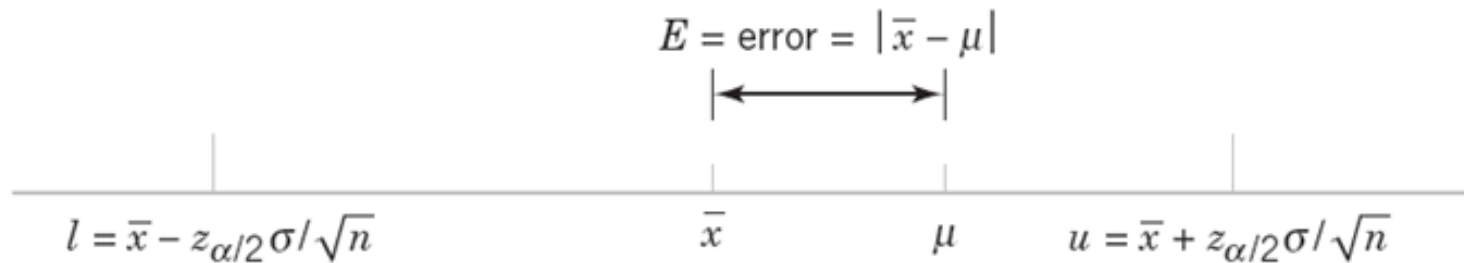
- The resulting 95% CI is found from:

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- **Practical Interpretation:** Based on the sample data, a range of highly plausible values for mean impact energy for A238 steel at 60°C is

Confidence Interval on the Mean of a Normal Distribution, Variance Known

- Confidence Level and Precision of Error
 - The length of a confidence interval is a measure of the **precision** of estimation.



Error in estimating μ with $\bar{x}$.

Confidence Interval on the Mean of a Normal Distribution, Variance Known

- Choice of Sample Size
- Definition

If $\bar{x}$ is used as an estimate of μ , we can be $100(1 - \alpha)\%$ confident that the error $|\bar{x} - \mu|$ will not exceed a specified amount E when the sample size is

$$n = \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2$$

Example: Metallic Material Transition

- Suppose that we wanted to determine **how many specimens must be tested** to ensure that the 95% CI on μ for A238 steel cut at 60°C has a length of at most 1.0 J.
- Since the bound on error in estimation E is one-half of the length of the CI, to determine n we use the equation with $E = 0.5$, $\sigma = 1$, and $z_{\alpha/2} = 1.96$.
- The required sample size is 16,

$$n = \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2 =$$

and because n must be an integer, the required sample size is $n=16$.

Confidence Interval on the Mean of a Normal Distribution, Variance Known

- One-Sided Confidence Bounds
- **Definition**

A $100(1 - \alpha)\%$ **upper-confidence bound** for μ is

$$\mu \leq u = \bar{x} + z_{\alpha}\sigma/\sqrt{n}$$

and a $100(1 - \alpha)\%$ **lower-confidence bound** for μ is

$$\bar{x} - z_{\alpha}\sigma/\sqrt{n} = l \leq \mu$$

Example: Metallic Material Transition

- Assume that impact energy is normally distributed with $\sigma = 1$ J.
- We want to find a 95% upper confidence bound and lower confidence bound for μ .
- The required quantities are $z_{\alpha} = z_{0.05} =$?, $n = 10$, $\sigma = 1$, and $\bar{x} = 64.46$.
- The resulting 95% upper confidence bound is: ...
- The resulting 95% lower confidence bound is: ...

Confidence Interval on the Mean of a Normal Distribution, Variance Known

- A Large-Sample Confidence Interval for μ
- **Definition**

When n is large, the quantity

$$\frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has an approximate standard normal distribution. Consequently,

$$\bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}}$$

is a **large sample confidence interval** for μ , with confidence level of approximately $100(1 - \alpha)\%$.

Confidence Interval on the Mean of a Normal Distribution, Variance Known

- A General Large Sample Confidence Interval

$$\hat{\theta} - z_{\alpha/2} \sigma_{\hat{\theta}} \leq \theta \leq \hat{\theta} + z_{\alpha/2} \sigma_{\hat{\theta}}$$

Example: Mercury Contamination

An article in the 1993 volume of the *Transactions of the American Fisheries Society* reports the results of a study to investigate the mercury contamination in largemouth bass. A

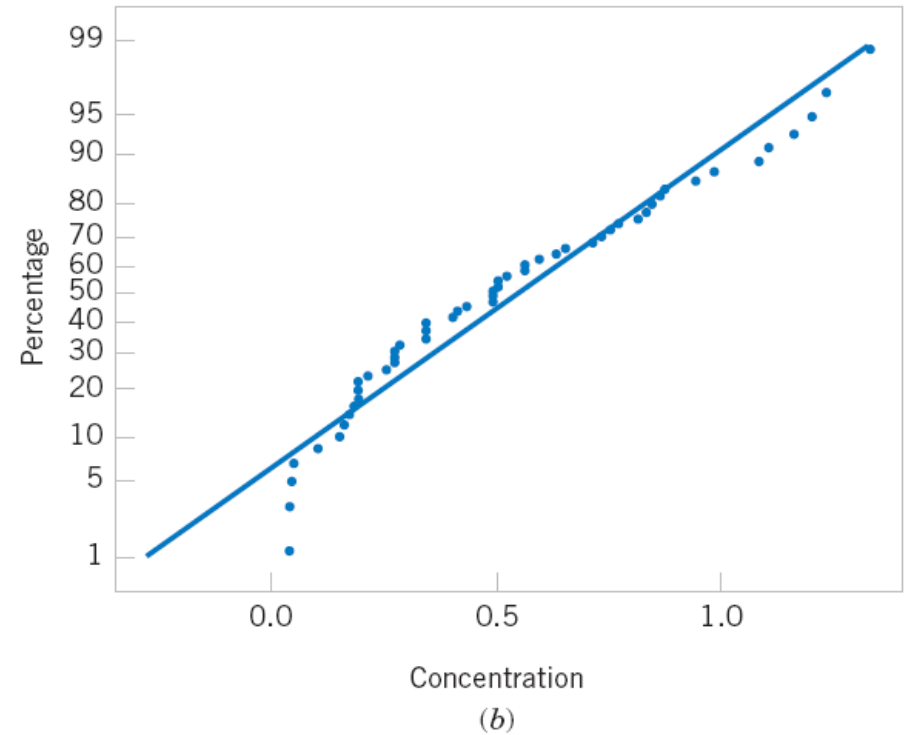
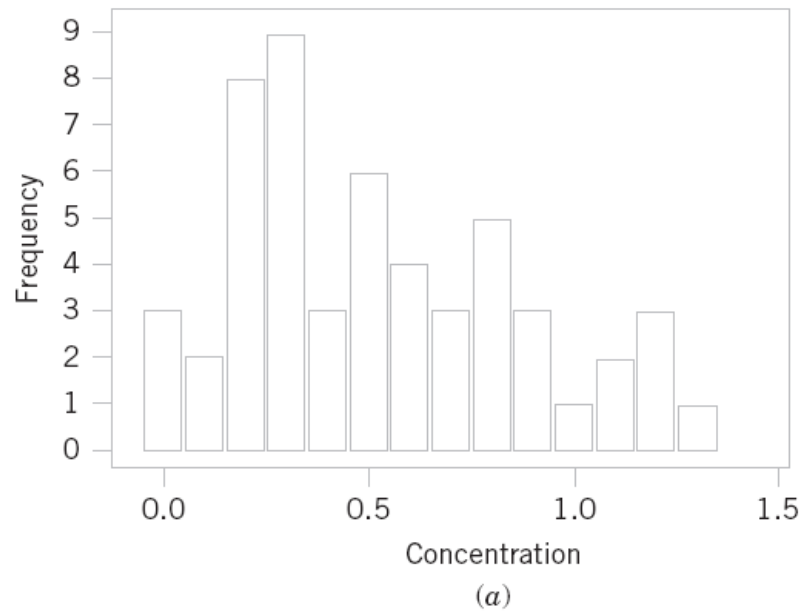
sample of fish was selected from 53 Florida lakes, and mercury concentration in the muscle tissue, was measured (ppm). The mercury concentration values are

1.230	0.490	0.490	1.080	0.590	0.280	0.180	0.100	0.940
1.330	0.190	1.160	0.980	0.340	0.340	0.190	0.210	0.400
0.040	0.830	0.050	0.630	0.340	0.750	0.040	0.860	0.430
0.044	0.810	0.150	0.560	0.840	0.870	0.490	0.520	0.250
1.200	0.710	0.190	0.410	0.500	0.560	1.100	0.650	0.270
0.270	0.500	0.770	0.730	0.340	0.170	0.160	0.270	

Descriptive Statistics: Concentration

Variable	N	Mean	Median	StDev
Concentration	53	0.5250	0.4900	0.3486
Variable	Minimum	Maximum	Q1	Q3
Concentration	0.0400	1.3300	0.2300	0.7900

Example: Mercury Contamination



Mercury concentration in largemouth bass (a) Histogram. (b) Normal probability plot

Decision Making for a Single Sample



- Statistical Inference and Point Estimation
- Inference on the Mean of a Population, Variance Known
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- Inference on the Variance of a Normal Population
- Inference on a Population Proportion
- Hypothesis Testing

Confidence Interval on the Mean of a Normal Distribution, Variance Unknown

- The t distribution

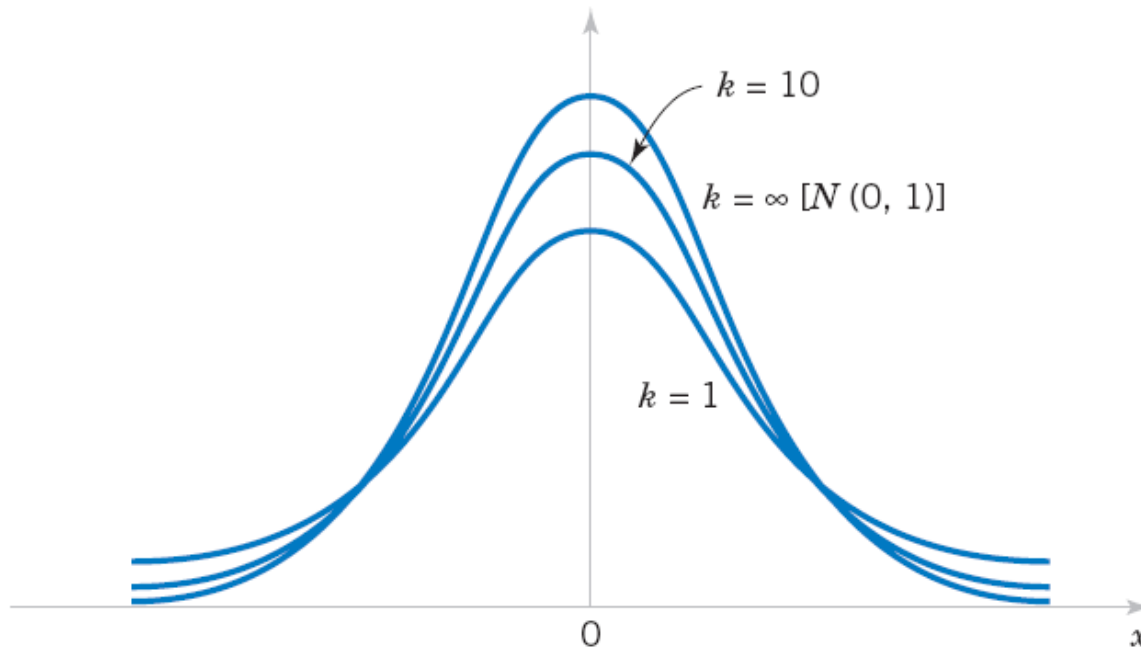
Let $X_1, X_2, \dots, X_n$ be a random sample from a normal distribution with unknown mean μ and unknown variance σ^2 . The random variable

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t distribution with $n - 1$ degrees of freedom.

Confidence Interval on the Mean of a Normal Distribution, Variance Unknown

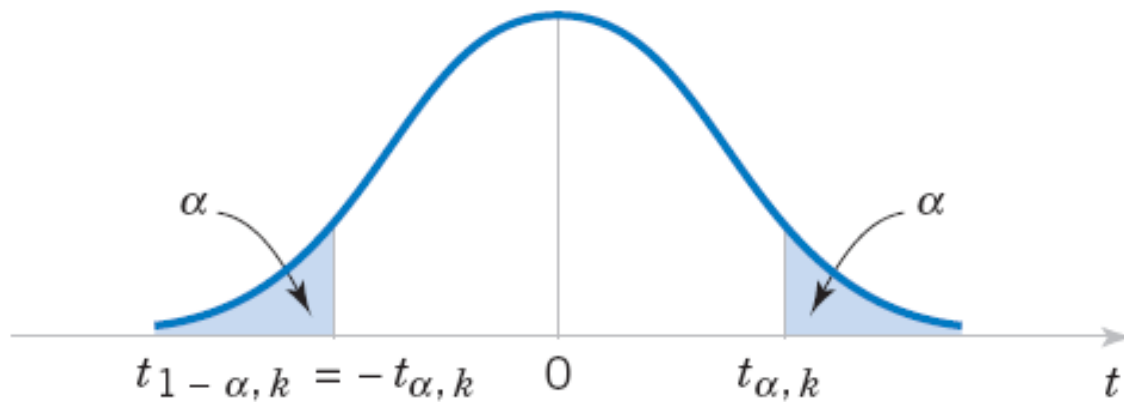
- The t distribution



Probability density functions of several t distributions.

Confidence Interval on the Mean of a Normal Distribution, Variance Unknown

- The t distribution



Percentage points of the t distribution.

Confidence Interval on the Mean of a Normal Distribution, Variance Unknown

- The t Confidence Interval on μ

If $\bar{x}$ and s are the mean and standard deviation of a random sample from a normal distribution with unknown variance σ^2 , a **100(1 - α)% confidence interval on μ** is given by

$$\bar{x} - t_{\alpha/2, n-1}s/\sqrt{n} \leq \mu \leq \bar{x} + t_{\alpha/2, n-1}s/\sqrt{n}$$

where $t_{\alpha/2, n-1}$ is the upper 100 α /2 percentage point of the t distribution with $n - 1$ degrees of freedom.

- **One-sided confidence bounds** on the mean are found by replacing $t_{\alpha/2, n-1}$ with $t_{\alpha, n-1}$.

Example

Alloy Adhesion

An article in the journal *Materials Engineering* (1989, Vol. II, No. 4, pp. 275–281) describes the results of tensile adhesion tests on 22 U-700 alloy specimens. The load at specimen failure is as follows (in megapascals):

19.8	10.1	14.9	7.5	15.4	15.4
15.4	18.5	7.9	12.7	11.9	11.4
11.4	14.1	17.6	16.7	15.8	
19.5	8.8	13.6	11.9	11.4	

The sample mean is $\bar{x} = 13.71$, and the sample standard deviation is $s = 3.55$. Figures 8-6 and 8-7 show a box plot and a normal probability plot of the tensile adhesion test data, respectively. These displays provide good support for the

assumption that the population is normally distributed. We want to find a 95% CI on μ . Since $n = 22$, we have $n - 1 = 21$ degrees of freedom for t , so $t_{0.025,21} = 2.080$. The resulting CI is

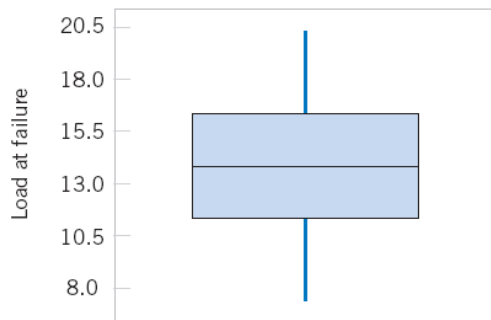


Figure 8-6 Box and whisker plot for the load at failure data in Example 8-5.

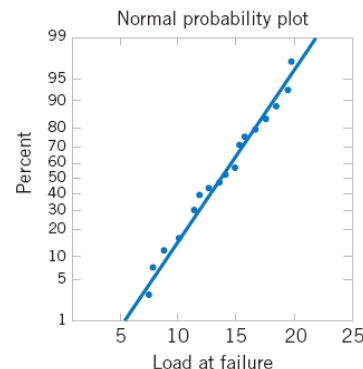


Figure 8-7 Normal probability plot of the load at failure data from Example 8-5.

Decision Making for a Single Sample



- Statistical Inference and Point Estimation
- Inference on the Mean of a Population, Variance Known
- Inference on the Mean of a Population, Variance Unknown
- Inference on the Variance of a Normal Population
- Inference on a Population Proportion
- Hypothesis Testing

Confidence Interval on the Variance and Standard Deviation of a Normal Distribution

- Definition

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal distribution with mean μ and variance σ^2 , and let S^2 be the sample variance. Then the random variable

$$X^2 = \frac{(n - 1) S^2}{\sigma^2}$$

has a chi-square (χ^2) distribution with $n - 1$ degrees of freedom.

Confidence Interval on the Variance and Standard Deviation of a Normal Distribution

- **Definition**

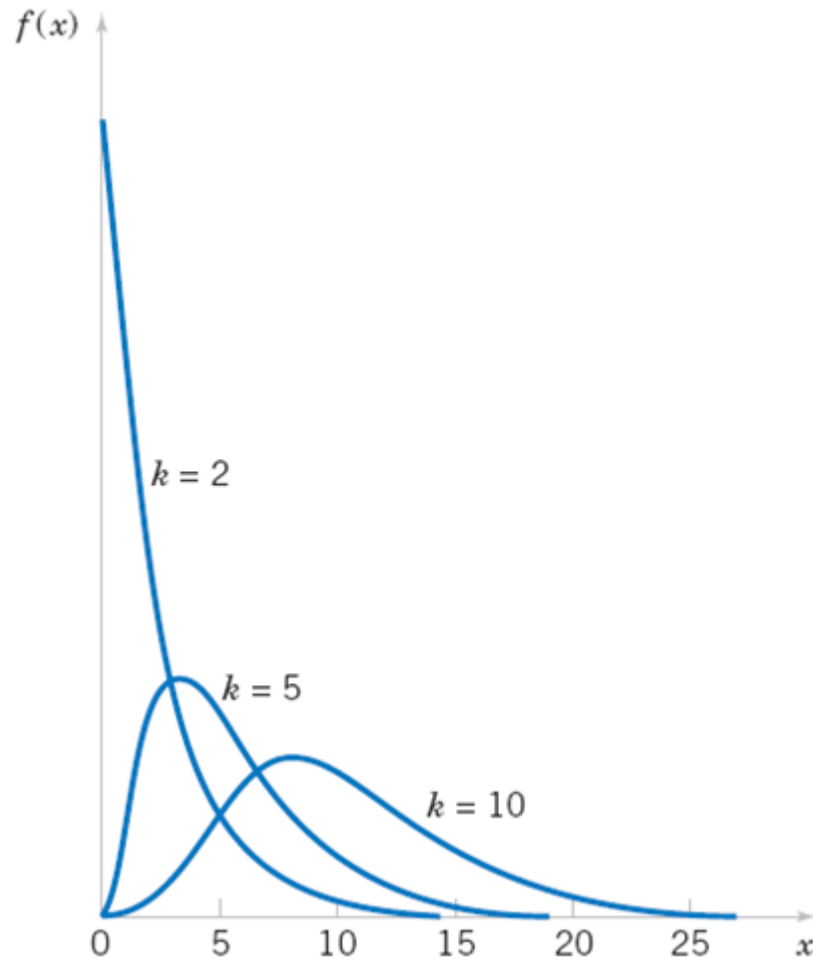
If s^2 is the sample variance from a random sample of n observations from a normal distribution with unknown variance σ^2 , then a **100(1 - α)% confidence interval on σ^2** is

$$\frac{(n - 1)s^2}{\chi_{\alpha/2, n-1}^2} \leq \sigma^2 \leq \frac{(n - 1)s^2}{\chi_{1-\alpha/2, n-1}^2}$$

where $\chi_{\alpha/2, n-1}^2$ and $\chi_{1-\alpha/2, n-1}^2$ are the upper and lower 100 $\alpha/2$ percentage points of the chi-square distribution with $n - 1$ degrees of freedom, respectively. A **confidence interval for σ** has lower and upper limits that are the square roots of the corresponding limits in this equation.

Confidence Interval on the Variance and Standard Deviation of a Normal Distribution

Probability density functions of several χ^2 distributions.



Confidence Interval on the Variance and Standard Deviation of a Normal Distribution

- One-Sided Confidence Bounds

The $100(1 - \alpha)\%$ lower and upper confidence bounds on σ^2 are

$$\frac{(n - 1)s^2}{\chi_{\alpha, n-1}^2} \leq \sigma^2 \quad \text{and} \quad \sigma^2 \leq \frac{(n - 1)s^2}{\chi_{1-\alpha, n-1}^2}$$

respectively.

Example

Detergent Filling

An automatic filling machine is used to fill bottles with liquid detergent. A random sample of 20 bottles results in a sample variance of fill volume of $s^2 = 0.0153$ (fluid ounce)². If the variance of fill volume is too large, an unacceptable proportion of bottles will be under- or overfilled. We will assume that the fill volume is approximately normally distributed. A 95% upper confidence bound is:



Decision Making for a Single Sample



- Statistical Inference and Point Estimation
- Inference on the Mean of a Population, Variance Known
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- Inference on a Population Proportion
- Hypothesis Testing

A Large-Sample Confidence Interval For a Population Proportion

- Normal Approximation for Binomial Proportion

If n is large, the distribution of

$$Z = \frac{X - np}{\sqrt{np(1-p)}} = \frac{\hat{P} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

is approximately standard normal.

- The quantity $\sqrt{p(1-p)/n}$ is called the standard error of the point estimator $\hat{p}$.

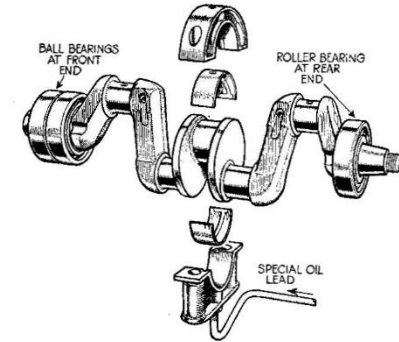
A Large-Sample Confidence Interval For a Population Proportion

If $\hat{p}$ is the proportion of observations in a random sample of size n that belongs to a class of interest, an approximate $100(1 - \alpha)\%$ confidence interval on the proportion p of the population that belongs to this class is

$$\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \leq p \leq \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution.

Example



Crankshaft Bearings

In a random sample of 85 automobile engine crankshaft bearings, 10 have a surface finish that is rougher than the specifications allow. Therefore, a point estimate of the proportion of bearings in the population that exceeds the roughness specification is $\hat{p} = x/n = 10/85 = 0.12$. A 95% two-sided confidence interval for p is computed as

A Large-Sample Confidence Interval For a Population Proportion

- One-Sided Confidence Bounds

The approximate $100(1 - \alpha)\%$ lower and upper confidence bounds are

$$\hat{p} - z_{\alpha} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \leq p \quad \text{and} \quad p \leq \hat{p} + z_{\alpha} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

respectively.

A Large-Sample Confidence Interval For a Population Proportion

- Choice of Sample Size
 - The sample size for a specified value E is given by

$$n = \left(\frac{z_{\alpha/2}}{E} \right)^2 p(1 - p)$$

- An upper bound on n is given by

$$n = \left(\frac{z_{\alpha/2}}{E} \right)^2 (0.25)$$

Example

Crankshaft Bearings

Consider the situation in [Crankshaft Bearings](#). How large a sample is required if we want to be 95% confident that the error in using $\hat{p}$ to estimate p is less than 0.05? Using $\hat{p} = 0.12$ as an initial estimate of p , the required sample size is

Decision Making for a Single Sample



- Statistical Inference and Point Estimation
- Inference on the Mean of a Population, Variance Known
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- Inference on the Variance of a Normal Population
- Inference on a Population Proportion
- Hypothesis Testing

Hypothesis Testing

Statistical Hypotheses

- **Statistical hypothesis testing** and **confidence interval estimation** of parameters are the fundamental methods used at the data analysis stage of a comparative experiment, in which the engineer is interested, for example, in comparing the mean of a population to a specified value.
- **Definition**

A **statistical hypothesis** is a statement about the parameters of one or more populations.

Hypothesis Testing

Statistical Hypotheses

- For example, suppose that we are interested in the **burning rate** of a solid propellant used to power aircrew escape systems.
 - Now burning rate is a **random variable** that can be described by a probability distribution.
 - Suppose that our interest focuses on **the mean burning rate** (a parameter of this distribution).
 - Specifically, we are interested in deciding whether or not the mean burning rate is 50 centimeters per second.



Statistical Hypotheses

- Two-sided Alternative Hypothesis

$H_0: \mu = 50$ centimeters per second

null hypothesis

$H_1: \mu \neq 50$ centimeters per second

alternative hypothesis

- One-sided Alternative Hypotheses

$H_0: \mu = 50$ centimeters per second

$H_0: \mu = 50$ centimeters per second

or

$H_1: \mu < 50$ centimeters per second

$H_1: \mu > 50$ centimeters per second

Statistical Hypotheses

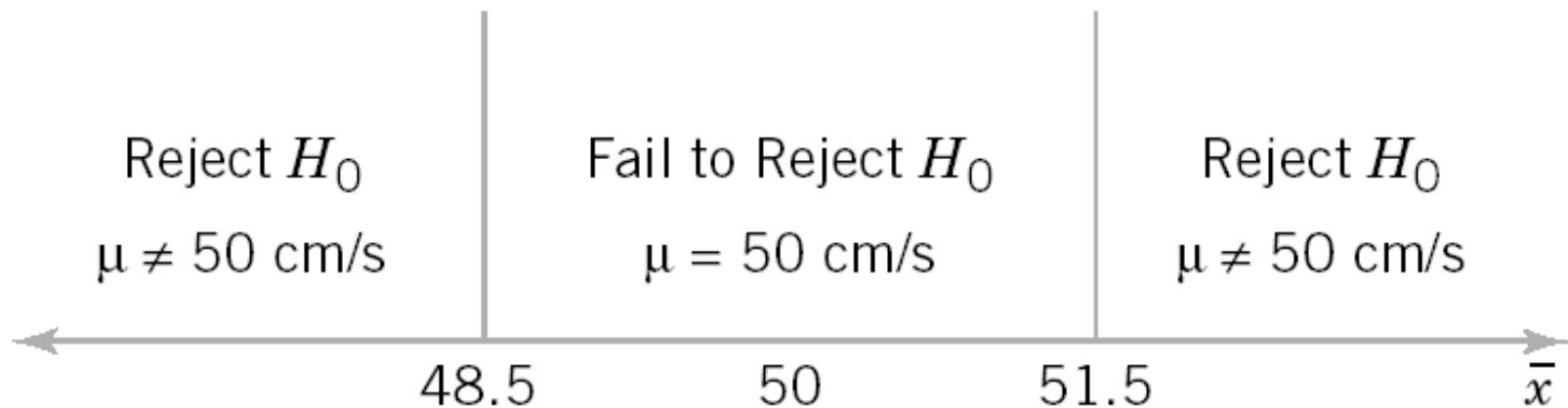
Test of a Hypothesis

- A procedure leading to a decision about a particular hypothesis
- Hypothesis-testing procedures rely on using the information in a **random sample from the population of interest**.
- If this information is *consistent* with the hypothesis, then we will conclude that the hypothesis is **true**; if this information is *inconsistent* with the hypothesis, we will conclude that the hypothesis is **false**.

Tests of Statistical Hypotheses

$H_0: \mu = 50$ centimeters per second

$H_1: \mu \neq 50$ centimeters per second



Decision criteria for testing $H_0: \mu = 50$ centimeters per second versus $H_1: \mu \neq 50$ centimeters per second.

Tests of Statistical Hypotheses

- Definitions

Rejecting the null hypothesis H_0 when it is true is defined as a **type I error**.

Failing to reject the null hypothesis when it is false is defined as a **type II error**.

Tests of Statistical Hypotheses

Decisions in Hypothesis Testing

Decision	H_0 Is True	H_0 Is False
Fail to reject H_0	no error	type II error
Reject H_0	type I error	no error

$$\alpha = P(\text{type I error}) = P(\text{reject } H_0 \text{ when } H_0 \text{ is true})$$

Sometimes the type I error probability is called the **significance level**, or the **α -error**, or the **size** of the test.

Tests of Statistical Hypotheses

$$\alpha = P(\bar{X} < 48.5 \text{ when } \mu = 50) + P(\bar{X} > 51.5 \text{ when } \mu = 50)$$

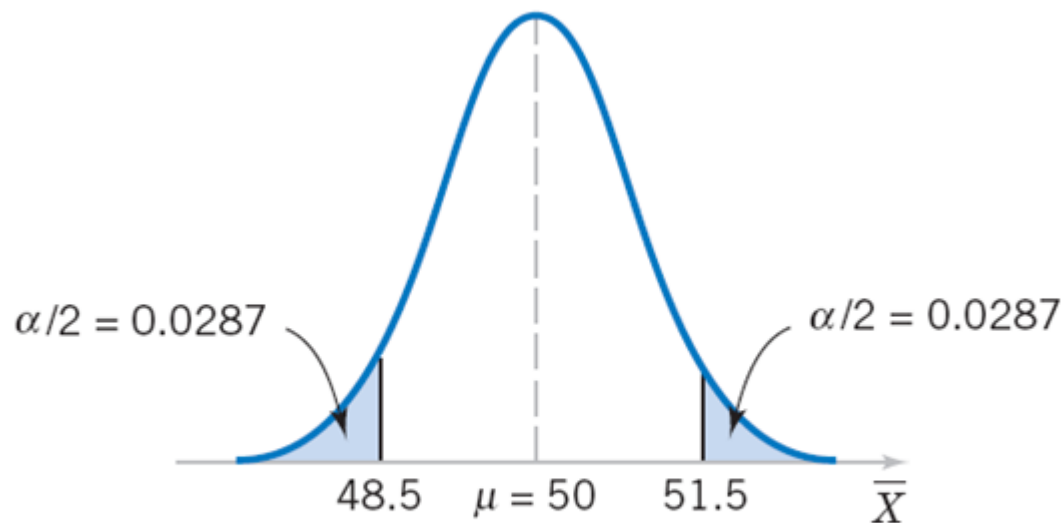
The z-values that correspond to the critical values 48.5 and 51.5 are

$$z_1 = \frac{48.5 - 50}{0.79} = -1.90 \quad \text{and} \quad z_2 = \frac{51.5 - 50}{0.79} = 1.90$$

Therefore

$$\alpha = P(Z < -1.90) + P(Z > 1.90) =$$

Tests of Statistical Hypotheses

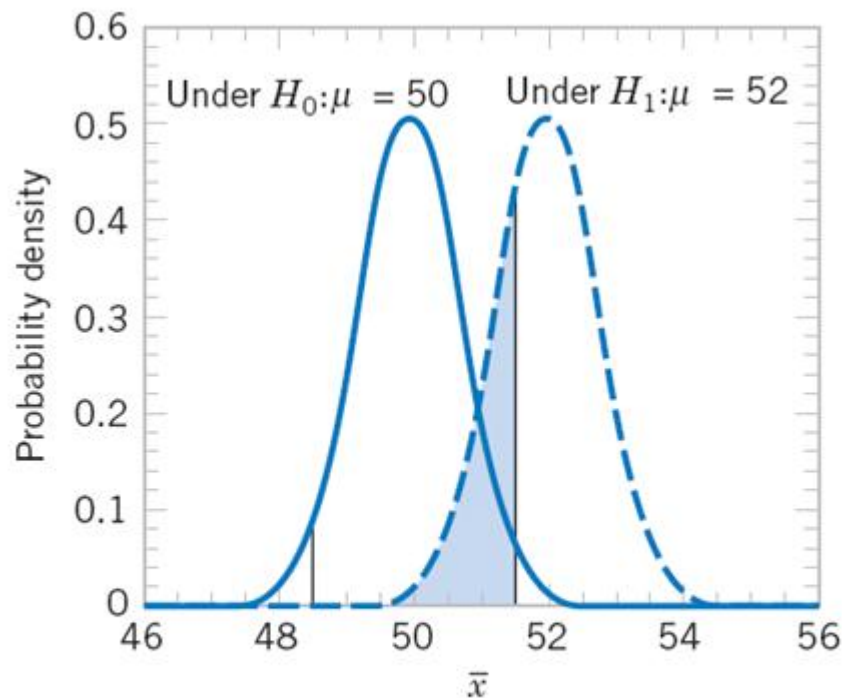


The critical region for $H_0: \mu = 50$ versus $H_1: \mu \neq 50$
and $n = 10$.

$$\alpha = P(\text{type I error}) = P(\text{reject } H_0 \text{ when } H_0 \text{ is true})$$

Tests of Statistical Hypotheses

$$\beta = P(\text{type II error}) = P(\text{fail to reject } H_0 \text{ when } H_0 \text{ is false})$$



The probability of
type II error when
 $\mu = 52$ and $n = 10$.

The probability of type II error when $\mu = 52$
and $n = 10$.

Tests of Statistical Hypotheses

$$\beta = P(48.5 \leq \bar{X} \leq 51.5 \text{ when } \mu = 52)$$

The z-values corresponding to 48.5 and 51.5 when $\mu = 52$ are

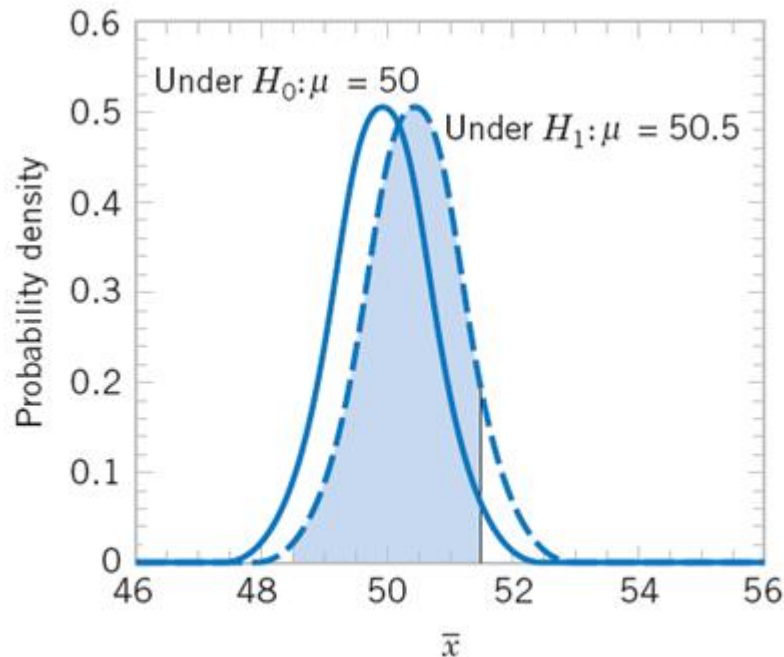
$$z_1 = \frac{48.5 - 52}{0.79} = -4.43 \quad \text{and} \quad z_2 = \frac{51.5 - 52}{0.79} = -0.63$$

Therefore

$$\beta = P(-4.43 \leq Z \leq -0.63) = P(Z \leq -0.63) - P(Z \leq -4.43)$$

Tests of Statistical Hypotheses

$$\beta = P(48.5 \leq \bar{X} \leq 51.5 \text{ when } \mu = 50.5)$$



The probability of type II error when $\mu = 50.5$ and $n = 10$.

The probability of type II error when $\mu = 50.5$ and $n = 10$.

Tests of Statistical Hypotheses

$$\beta = P(48.5 \leq \bar{X} \leq 51.5 \text{ when } \mu = 50.5)$$

The z-values corresponding to 48.5 and 51.5 when $\mu = 50.5$ are

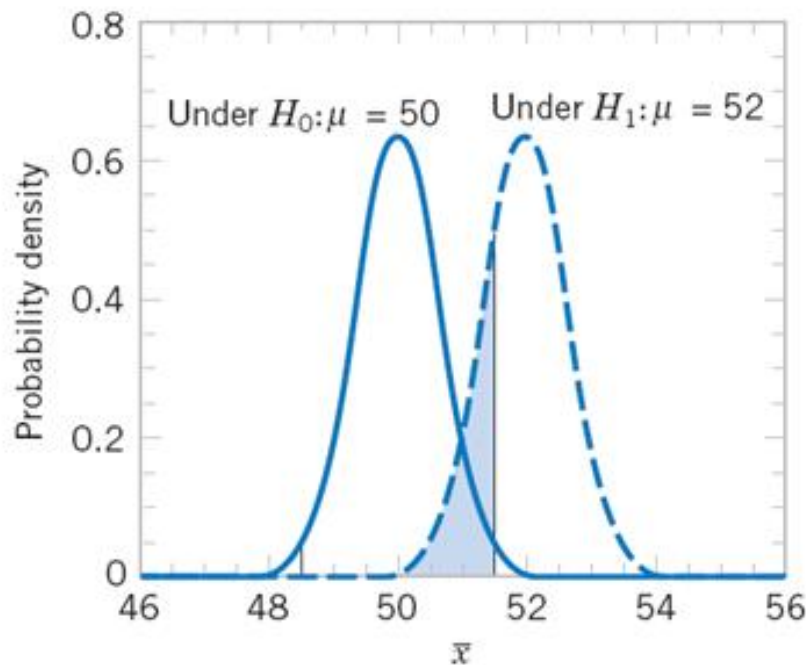
$$z_1 = \frac{48.5 - 50.5}{0.79} = -2.53 \quad \text{and} \quad z_2 = \frac{51.5 - 50.5}{0.79} = 1.27$$

Therefore

$$\beta = P(-2.53 \leq Z \leq 1.27) = P(Z \leq 1.27) - P(Z \leq -2.53)$$

Tests of Statistical Hypotheses

$$\beta = P(48.5 \leq \bar{X} \leq 51.5 \text{ when } \mu = 52)$$



The probability of
type II error when
 $\mu = 52$ and $n = 16$.

Tests of Statistical Hypotheses

$$\beta = P(48.5 \leq \bar{X} \leq 51.5 \text{ when } \mu = 52)$$

When $n = 16$, the standard deviation of $\bar{X}$ is $\sigma/\sqrt{n} = 2.5/\sqrt{16} = 0.625$, and the z -values corresponding to 48.5 and 51.5 when $\mu = 52$ are

$$z_1 = \frac{48.5 - 52}{0.625} = -5.60 \quad \text{and} \quad z_2 = \frac{51.5 - 52}{0.625} = -0.80$$

Therefore

$$\beta = P(-5.60 \leq Z \leq -0.80) = P(Z \leq -0.80) - P(Z \leq -5.60)$$

Hypothesis Testing

Acceptance Region	Sample Size	α	β at $\mu = 52$	β at $\mu = 50.5$
$48.5 < \bar{x} < 51.5$	10	0.0576	0.2643	0.8923
$48 < \bar{x} < 52$	10	0.0114	0.5000	0.9705
$48.81 < \bar{x} < 51.19$	16	0.0576	0.0966	0.8606
$48.42 < \bar{x} < 51.58$	16	0.0114	0.2515	0.9578

Hypothesis Testing

- Definition

The **power** of a statistical test is the probability of rejecting the null hypothesis H_0 when the alternative hypothesis is true.

- The power is computed as $1 - \beta$, and power can be interpreted as *the probability of correctly rejecting a false null hypothesis*. We often compare statistical tests by comparing their power properties.
- For example, consider the propellant burning rate problem when we are testing $H_0 : m = 50$ centimeters per second against $H_1 : m$ not equal 50 centimeters per second. Suppose that the true value of the mean is $m = 52$. When $n = 10$, we found that $\beta = 0.2643$, so the power of this test is $1 - \beta = 1 - 0.2643 = 0.7357$ when $m = 52$.

One-Sided and Two-Sided Hypotheses

- Two-Sided Test:

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0$$

- One-Sided Tests:

$$\begin{array}{ll} H_0: \mu = \mu_0 & \text{or} \\ H_1: \mu > \mu_0 & H_0: \mu = \mu_0 \\ & H_1: \mu < \mu_0 \end{array}$$

Example

Propellant Burning Rate

Consider the propellant burning rate problem. Suppose that if the burning rate is less than 50 centimeters per second, we wish to show this with a strong conclusion. The hypotheses should be stated as

$$H_0: \mu = 50 \text{ centimeters per second}$$

$$H_1: \mu < 50 \text{ centimeters per second}$$

Here the critical region lies in the lower tail of the distribution of $\bar{X}$. Since the rejection of H_0 is always a strong conclusion, this statement of the hypotheses will produce the desired outcome if H_0 is rejected. Notice that, although the null hypothesis is stated with an equals sign, it is understood to include any value of μ not specified by the alternative hypothesis. Therefore, failing to reject H_0 does not mean that $\mu = 50$ centimeters per second exactly, but only that we do not have strong evidence in support of H_1 .

P-Values in Hypothesis Tests

- Definition

The **P-value** is the smallest level of significance that would lead to rejection of the null hypothesis H_0 with the given data.

Consider the two-sided hypothesis test for burning rate

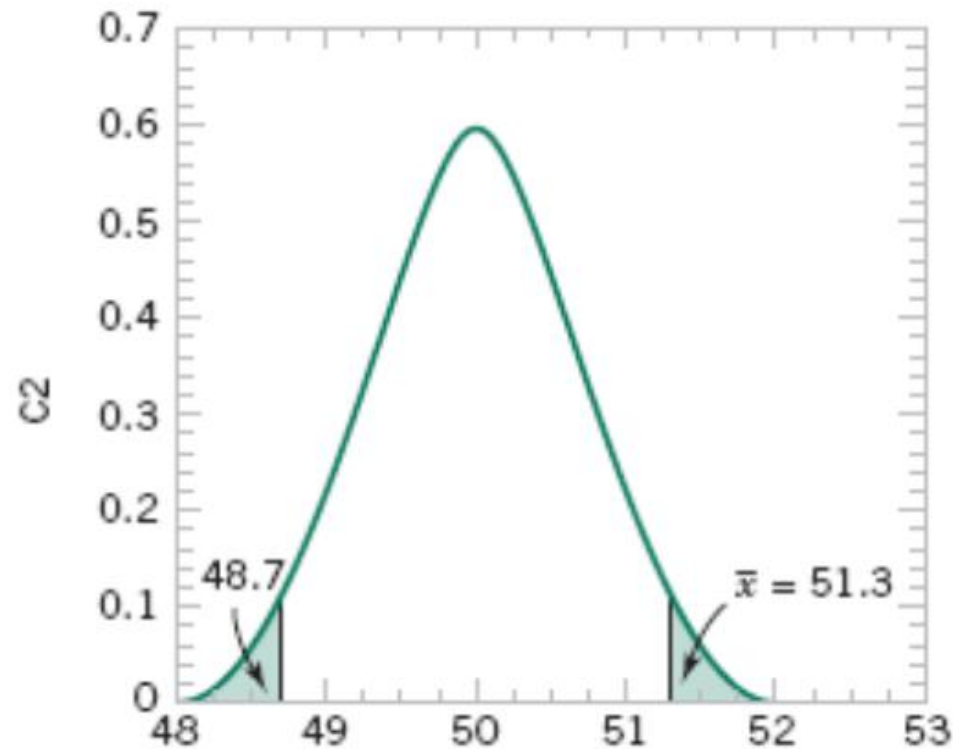
$$H_0 : \mu = 50 \quad H_1 : \mu \neq 50$$

with $n = 16$ and $\sigma = 2.5$. Suppose that the observed sample mean is $\bar{x} = 51.3$ centimeters per second. A critical region for this test with critical values at 51.3 and the symmetric value 48.7. The P -value of the test is the α associated with this critical region. Any smaller value for α expands the critical region and the test fails to reject the null hypothesis when $\bar{x} = 51.3$. The P -value is easy to compute after the test statistic is observed.

$$\begin{aligned} P\text{-value} &= 1 - P(48.7 < \bar{X} < 51.3) \\ &= 1 - P\left(\frac{48.7 - 50}{2.5/\sqrt{16}} < Z < \frac{51.3 - 50}{2.5/\sqrt{16}}\right) \end{aligned}$$

P-Values in Hypothesis Tests

P -value
is area of shaded
region when $\bar{x} = 51.3$.



Connection between Hypothesis Tests and Confidence Intervals

There is a close relationship between the test of a hypothesis about any parameter, say θ , and the confidence interval for θ . If $[l, u]$ is a $100(1 - \alpha)\%$ confidence interval for the parameter θ , the test of size α of the hypothesis

$$H_0: \theta = \theta_0$$

$$H_1: \theta \neq \theta_0$$

will lead to rejection of H_0 if and only if θ_0 is **not** in the $100(1 - \alpha)\%$ CI $[l, u]$. As an illustration, consider the escape system propellant problem with $\bar{x} = 51.3$, $\sigma = 2.5$, and $n = 16$. The null hypothesis $H_0: \mu = 50$ was rejected, using $\alpha = 0.05$. The 95% two-sided CI on μ can be calculated. This CI is $51.3 \pm 1.96(2.5/\sqrt{16})$ and this is $50.075 \leq \mu \leq 52.525$. Because the value $\mu_0 = 50$ is not included in this interval, the null hypothesis $H_0: \mu = 50$ is rejected.

General Procedure for Hypothesis Tests

1. From the problem context, identify the **parameter of interest**.
2. State the **null hypothesis**, H_0 .
3. Specify an appropriate **alternative hypothesis**, H_1 .
4. Choose a **significance level**, α .
5. Determine an appropriate **test statistic**.
6. State the **rejection region** for the statistic.
7. Compute any necessary sample quantities, substitute these into the equation for the test statistic, and **compute** that value.
8. Decide **whether or not H_0 should be rejected** and report that in the problem context.

Tests on the Mean of a Normal Distribution, Variance Known

Hypothesis Tests on the Mean

- We wish to test:

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0$$

- The test statistic is:

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

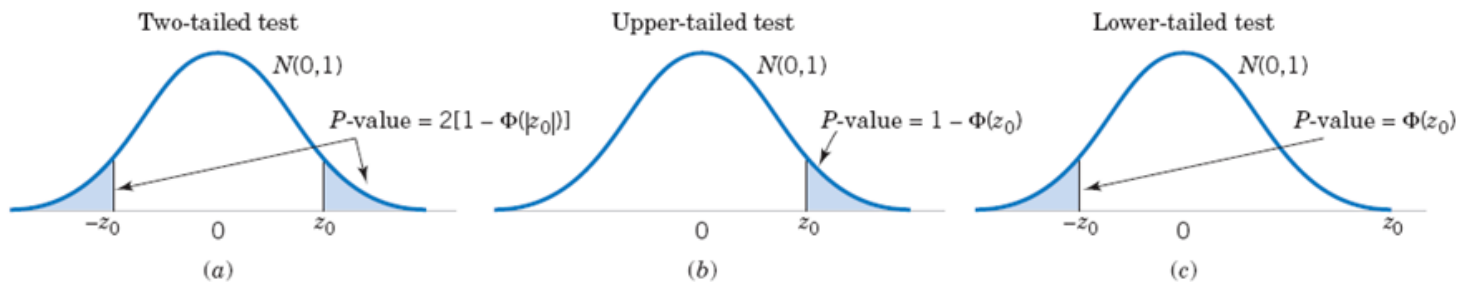
Hypothesis Tests on the Mean

- Reject H_0 if the observed value of the test statistic z_0 is either:

$$z_0 < -z_{\alpha/2} \text{ or } z_0 > z_{\alpha/2}$$

- Fail to reject H_0 if

$$-z_{\alpha/2} \leq z_0 \leq z_{\alpha/2}$$



The P -value for a z -test. (a) The two-sided alternative $H_1: \mu \neq \mu_0$. (b) The one-sided alternative $H_1: \mu > \mu_0$. (c) The one-sided alternative $H_1: \mu < \mu_0$.

Example

Aircrew escape systems are powered by a solid propellant. The burning rate of this propellant is an important product characteristic. Specifications require that the mean burning rate must be 50 centimeters per second. We know that the standard deviation of burning rate is $\sigma = 2$ centimeters per second. The experimenter decides to specify a type I error probability or significance level of $\alpha = 0.05$ and selects a random sample of $n = 25$ and obtains a sample average burning rate of $\bar{x} = 51.3$ centimeters per second. What conclusions should be drawn?

Solution

Propellant Burning Rate

Air crew escape systems are powered by a solid propellant. The burning rate of this propellant is an important product characteristic. Specifications require that the mean burning rate must be 50 centimeters per second. We know that the standard deviation of burning rate is $\sigma = 2$ centimeters per second. The experimenter decides to specify a type I error probability or significance level of $\alpha = 0.05$ and selects a random sample of $n = 25$ and obtains a sample average burning rate of $\bar{x} = 51.3$ centimeters per second. What conclusions should be drawn?

1. **Parameter of interest:** The parameter of interest is μ , the mean burning rate.
2. **Null hypothesis, H_0 :** $H_0: \mu = 50$ centimeters per second
3. **Alternative hypothesis, H_1 :** $H_1: \mu \neq 50$ centimeters per second

Solution

4. **Test statistic:** The test statistic is

$$z_0 = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

5. **Reject H_0 if:** Reject H_0 if the P -value is less than 0.05. To use a fixed significance level test, the boundaries of the critical region would be $z_{0.025} = 1.96$ and $-z_{0.025} = -1.96$.

6. **Computations:** Since $\bar{x} = 51.3$ and $\sigma = 2$,

$$z_0 =$$

7. **Conclusion:** Since $z_0 = 3.25$ the p -value is

$$P\text{-value} = 2[1 - \Phi(3.25)] =$$

we reject H_0 ; $\mu = 50$ at the 0.05 level of significance.

Practical Interpretation: We conclude that the mean burning rate differs from 50 centimeters per second, based on a sample of 25 measurements. In fact, there is strong evidence that the mean burning rate exceeds 50 centimeters per second.

Hypothesis Tests on the Mean

Null hypothesis: $H_0: \mu = \mu_0$

Test statistic: $Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$

Alternative hypothesis	Rejection criteria
$H_1: \mu \neq \mu_0$	$z_0 > z_{\alpha/2, n-1}$ or $z_0 < -z_{\alpha/2, n-1}$
$H_1: \mu > \mu_0$	$z_0 > z_{\alpha, n-1}$
$H_1: \mu < \mu_0$	$z_0 < -z_{\alpha, n-1}$

The **P-value** is the smallest level of significance that would lead to rejection of the null hypothesis H_0 with the given data.

$$P = \begin{cases} 2[1 - \Phi(|z_0|)] & \text{for a two-tailed test: } H_0: \mu = \mu_0 & H_1: \mu \neq \mu_0 \\ 1 - \Phi(z_0) & \text{for an upper-tailed test: } H_0: \mu = \mu_0 & H_1: \mu > \mu_0 \\ \Phi(z_0) & \text{for a lower-tailed test: } H_0: \mu = \mu_0 & H_1: \mu < \mu_0 \end{cases}$$

Tests on the Mean of a Normal Distribution, Variance Unknown

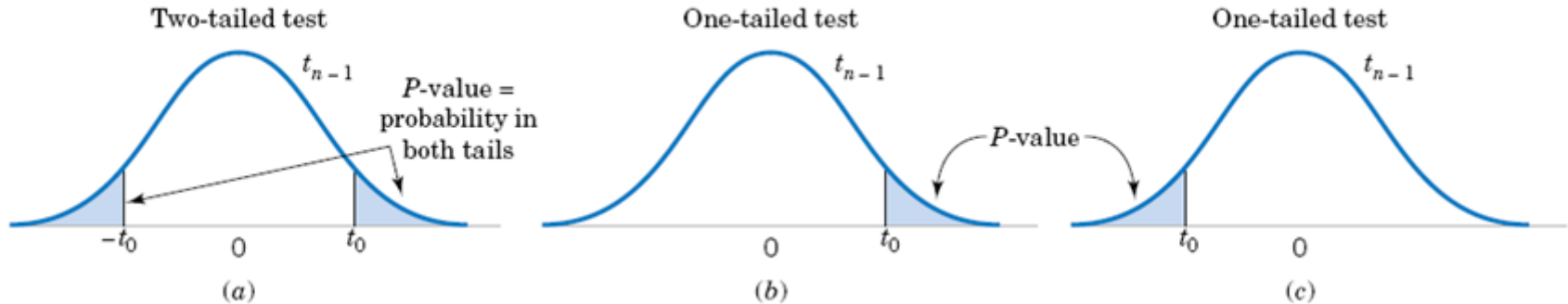
- One-Sample t -Test

Null hypothesis: $H_0: \mu = \mu_0$

Test statistic: $T_0 = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$

Alternative hypothesis	Rejection criteria
$H_1: \mu \neq \mu_0$	$t_0 > t_{\alpha/2, n-1}$ or $t_0 < -t_{\alpha/2, n-1}$
$H_1: \mu > \mu_0$	$t_0 > t_{\alpha, n-1}$
$H_1: \mu < \mu_0$	$t_0 < -t_{\alpha, n-1}$

Tests on the Mean of a Normal Distribution, Variance Unknown



The reference distribution for $H_0: \mu = \mu_0$ with critical region for (a) $H_1: \mu \neq \mu_0$, (b) $H_1: \mu > \mu_0$, and (c) $H_1: \mu < \mu_0$.

Example: Golf Club Design

The increased availability of light materials with high strength has revolutionized the design and manufacture of golf clubs, particularly drivers. Clubs with hollow heads and very thin faces can result in much longer tee shots, especially for players of modest skills. This is due partly to the “spring-like effect” that the thin face imparts to the ball. Firing a golf ball at the head of the club and measuring the ratio of the outgoing velocity of the ball to the incoming velocity can quantify this spring-like effect. The ratio of velocities is called the coefficient of restitution of the club. An experiment was performed in which 15 drivers produced by a particular club maker were selected at random and their coefficients of restitution measured. In the experiment the golf balls were fired from an air cannon so that the incoming velocity and spin rate of the ball could be precisely controlled. It is of interest to determine if there is evidence (with $\alpha = 0.05$) to support a claim that the mean coefficient of restitution exceeds 0.82. The observations follow:

0.8411	0.8191	0.8182	0.8125	0.8750
0.8580	0.8532	0.8483	0.8276	0.7983
0.8042	0.8730	0.8282	0.8359	0.8660

The sample mean and sample standard deviation are $\bar{x} = 0.83725$ and $s = 0.02456$.



Example: Golf Club Design

The solution using the seven-step procedure for hypothesis testing is as follows:

1. **Parameter of interest:** The parameter of interest is the mean coefficient of restitution, μ .
2. **Null hypothesis:**
3. **Alternative hypothesis:**
4. **Test Statistic:** The test statistic is
5. **Reject H_0 if :**
6. **Computations:**
7. **Conclusions:** From Appendix A Table II we find, for a t distribution with 14 degrees of freedom, that $t_0 = 2.72$



SAT scores

- In a random sample of 18 high school juniors, the mean of scores on the verbal portion of the SAT scores was found to be 510 with a standard deviation for the sample of 140 points.
- The mean for these scores is believed to be 500. Test the claim that the sample mean is not significantly different from the value 500. State the null hypothesis, test statistic, critical value for alpha of 0.01, conclusion, and interpretation.

Hypothesis Tests on the Variance and Standard Deviation of a Normal Distribution

Hypothesis Test on the Variance

Suppose that we wish to test the hypothesis that the variance of a normal population σ^2 equals a specified value, say σ_0^2 , or equivalently, that the standard deviation σ is equal to σ_0 . Let $X_1, X_2, \dots, X_n$ be a random sample of n observations from this population. To test

$$H_0: \sigma^2 = \sigma_0^2$$

$$H_1: \sigma^2 \neq \sigma_0^2$$

we will use the test statistic:

$$X_0^2 = \frac{(n-1)S^2}{\sigma_0^2}$$

Hypothesis Tests on the Variance and Standard Deviation of a Normal Distribution

Hypothesis Test on the Variance

If the null hypothesis $H_0: \sigma^2 = \sigma_0^2$ is true, the test statistic X_0^2 defined follows the chi-square distribution with $n - 1$ degrees of freedom. This is the reference distribution for this test procedure. Therefore, we calculate χ_0^2 , the value of the test statistic X_0^2 , and the null hypothesis $H_0: \sigma^2 = \sigma_0^2$ would be rejected if

$$\chi_0^2 > \chi_{\alpha/2, n-1}^2 \quad \text{or if} \quad \chi_0^2 < \chi_{1-\alpha/2, n-1}^2$$

where $\chi_{\alpha/2, n-1}^2$ and $\chi_{1-\alpha/2, n-1}^2$ are the upper and lower $100\alpha/2$ percentage points of the chi-square distribution with $n - 1$ degrees of freedom, respectively.

Hypothesis Tests on the Variance and Standard Deviation of a Normal Distribution

Hypothesis Test on the Variance

The same test statistic is used for one-sided alternative hypotheses. For the one-sided hypotheses

$$H_0: \sigma^2 = \sigma_0^2$$

$$H_1: \sigma^2 > \sigma_0^2$$

we would reject H_0 if $\chi_0^2 > \chi_{\alpha, n-1}^2$, whereas for the other one-sided hypotheses

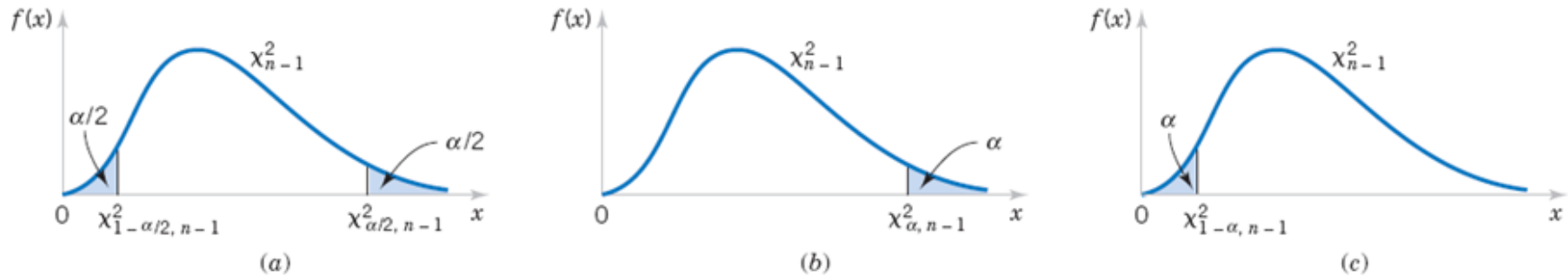
$$H_0: \sigma^2 = \sigma_0^2$$

$$H_1: \sigma^2 < \sigma_0^2$$

we would reject H_0 if $\chi_0^2 < \chi_{1-\alpha, n-1}^2$.

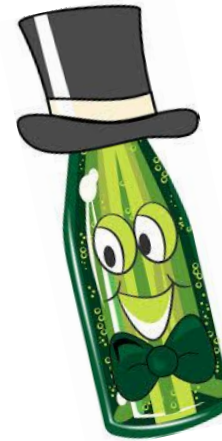
Hypothesis Tests on the Variance and Standard Deviation of a Normal Distribution

Hypothesis Test on the Variance



Reference distribution for the test of $H_0: \sigma^2 = \sigma_0^2$ with critical region values for (a) $H_1: \sigma^2 \neq \sigma_0^2$, (b) $H_1: \sigma^2 > \sigma_0^2$, and (c) $H_1: \sigma^2 < \sigma_0^2$.

Example: Automated Filling



An automated filling machine is used to fill bottles with liquid detergent. A random sample of 20 bottles results in a sample variance of fill volume of $s^2 = 0.0153$ (fluid ounces)². If the variance of fill volume exceeds 0.01 (fluid ounces)², an unacceptable proportion of bottles will be underfilled or overfilled. Is there evidence in the sample data to suggest that the manufacturer has a problem with underfilled or overfilled bottles? Use $\alpha = 0.05$, and assume that fill volume has a normal distribution.

Example: Automated Filling

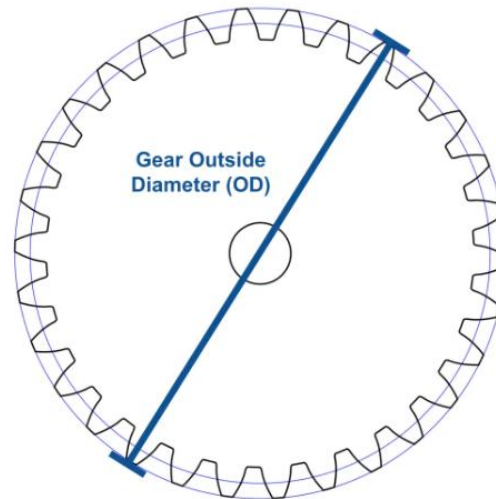
Using the seven-step procedure results in the following:

1. **Parameter of Interest:** The parameter of interest is the population variance σ^2 .



Gear Diameter

The observed variance for the 101 measurements of gear diameter is 0.00003969 (the standard deviation is 0.0063). We would like to test the null hypothesis that the true variance is equal to 0.01 at 95% CI.



Tests on a Population Proportion

Large-Sample Tests on a Proportion

- Many engineering decision problems include hypothesis testing about p .

$$H_0: p = p_0$$

$$H_1: p \neq p_0$$

- An appropriate **test statistic** is

$$Z_0 = \frac{X - np_0}{\sqrt{np_0(1 - p_0)}}$$

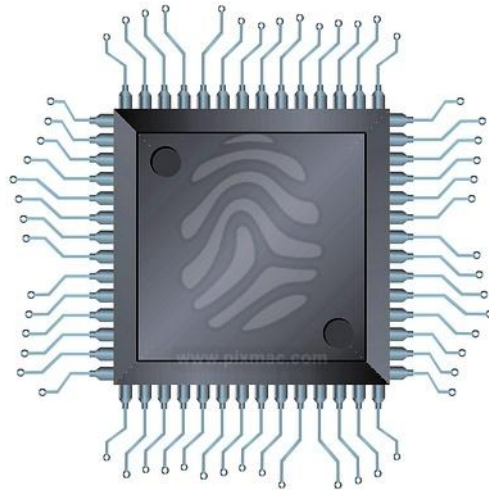
Tests on a Population Proportion

- Another form of the test statistic Z_0 is

$$Z_0 = \frac{X/n - p_0}{\sqrt{p_0(1 - p_0)/n}} \quad \text{or} \quad Z_0 = \frac{\hat{P} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

Example: Automobile Engine Controller

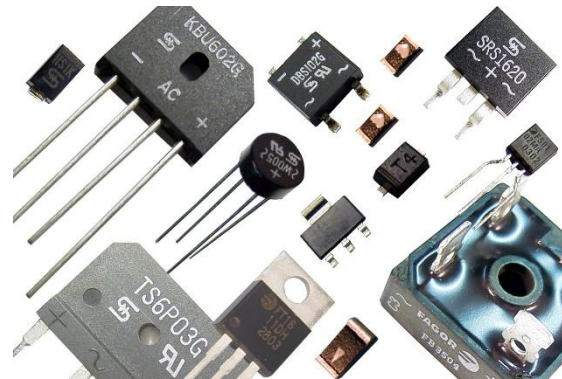
A semiconductor manufacturer produces controllers used in automobile engine applications. The customer requires that the process fallout or fraction defective at a critical manufacturing step not exceed 0.05 and that the manufacturer demonstrate process capability at this level of quality using $\alpha = 0.05$. The semiconductor manufacturer takes a random sample of 200 devices and finds that four of them are defective. Can the manufacturer demonstrate process capability for the customer?



Example: Automobile Engine Controller

We may solve this problem using the seven-step hypothesis-testing procedure as follows:

1. **Parameter of Interest:** The parameter of interest is the process fraction defective p .



Election

Suppose 60% of citizens voted in last election. 92 out of 148 people in a telephone survey said that they voted in current election.

At 0.05 significance level, can we reject the null hypothesis that the proportion of voters in the population is above 60% this year?